
Why Propagate Alone?

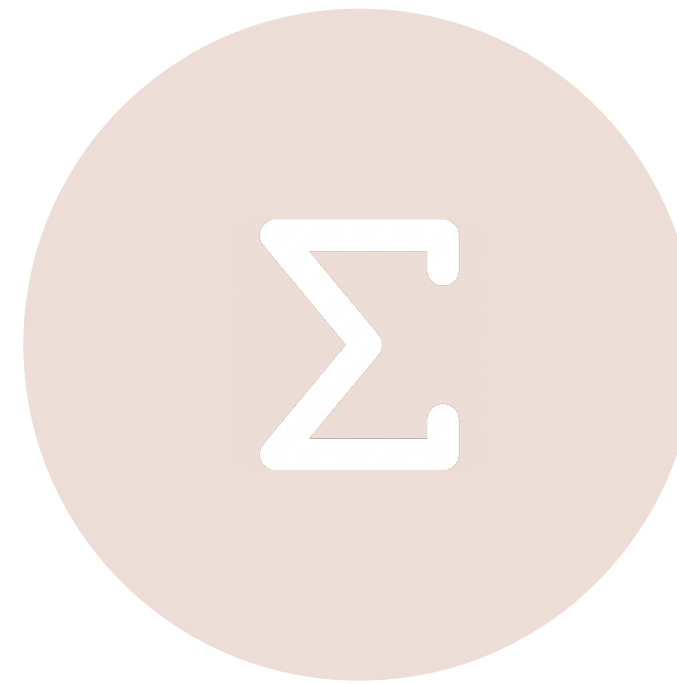
Parallel Use of Labels and Features on Graphs

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Outline



Background



Analysis

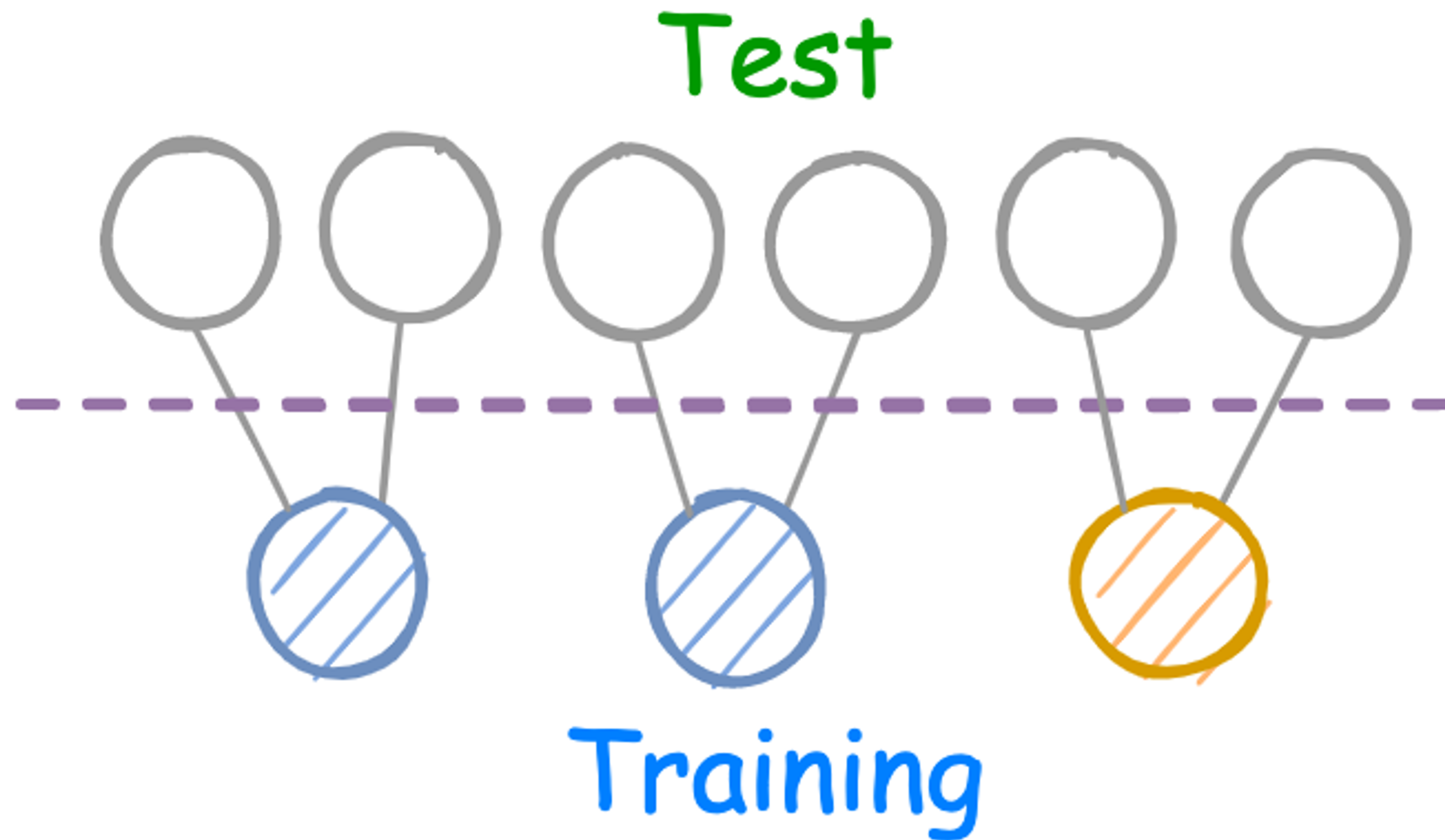


Applications

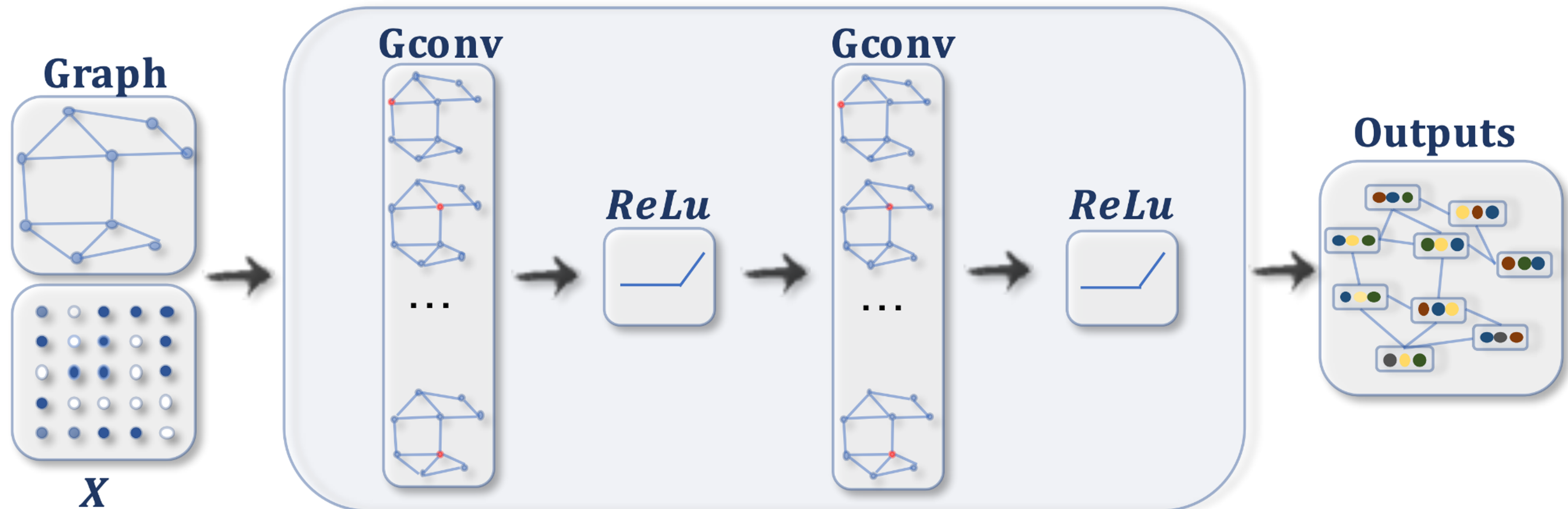


Background

Our Task: Node Classification

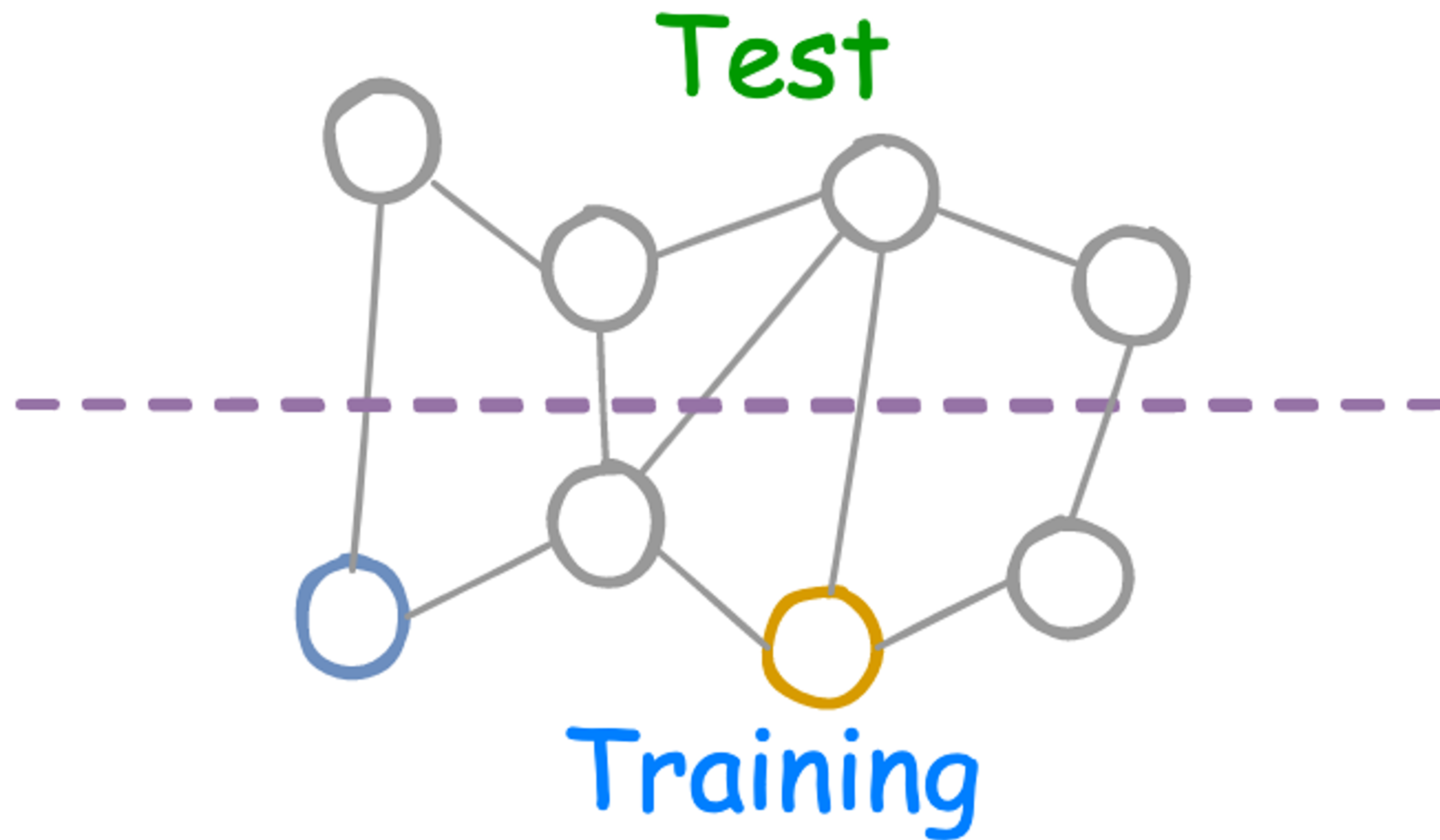


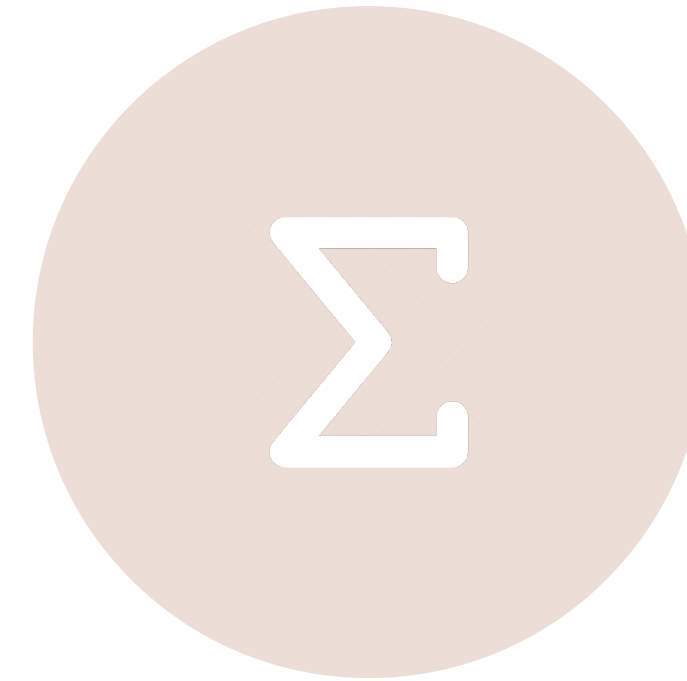
Graph Neural Networks for Node Classification



Training

Label Trick





Analysis

Optimization Objective

$$\mathbb{E}_{\text{plits}} \left[\sum_{i \in \mathcal{D}_{out}} \ell(\mathbf{y}_i, f[\mathbf{X}, \mathbf{Y}_{in}; \mathbf{w}]_i) \right]$$

Decompose the Objective

Theorem 1 Define $\Gamma = (\text{diag}(\mathbf{P}^T \mathbf{P}) - \mathbf{C}^T \mathbf{C})^{\frac{1}{2}} \mathbf{Y}_{tr}$. Then the label trick loss satisfies

$$\frac{1}{1 - \alpha} \mathbb{E}_{splits} \left[\|\mathbf{Y}_{out} - \mathbf{M}_{out} \mathbf{P} \tilde{\mathbf{Y}}_{in} \mathbf{W}\|_F^2 \right] = \|\mathbf{Y}_{tr} - \mathbf{M}_{tr} (\mathbf{P} - \mathbf{C}) \mathbf{Y}_{tr} \mathbf{W}\|_F^2 + \frac{1 - \alpha}{\alpha} \|\Gamma \mathbf{W}\|_F^2.$$

Corollary 1 Under the same conditions as Theorem 1, if we add the node features, we have that

$$\begin{aligned} & \frac{1}{1 - \alpha} \mathbb{E}_{splits} \left[\|\mathbf{Y}_{out} - \mathbf{M}_{out} \mathbf{P} \mathbf{X} \mathbf{W}_x - \mathbf{M}_{out} \mathbf{P} \tilde{\mathbf{Y}}_{in} \mathbf{W}_y\|_F^2 \right] \\ &= \|\mathbf{Y}_{tr} - \mathbf{M}_{tr} \mathbf{P} \mathbf{X} \mathbf{W}_x - \mathbf{M}_{tr} (\mathbf{P} - \mathbf{C}) \mathbf{Y}_{tr} \mathbf{W}_y\|_F^2 + \frac{1 - \alpha}{\alpha} \|\Gamma \mathbf{W}_y\|_F^2. \end{aligned} \tag{9}$$



data-fitting term



regularization term

Self-excluded Predictor from the Theorem

Corollary 1 *Under the same conditions as Theorem 1, if we add the node features, we have that*

$$\begin{aligned} & \frac{1}{1-\alpha} \mathbb{E}_{\text{splits}} \left[\| \mathbf{Y}_{out} - \mathbf{M}_{out} \mathbf{P} \mathbf{X} \mathbf{W}_x - \mathbf{M}_{out} \mathbf{P} \tilde{\mathbf{Y}}_{in} \mathbf{W}_y \|_F^2 \right] \\ &= \| \mathbf{Y}_{tr} - \mathbf{M}_{tr} \mathbf{P} \mathbf{X} \mathbf{W}_x - \mathbf{M}_{tr} (\mathbf{P} - \mathbf{C}) \mathbf{Y}_{tr} \mathbf{W}_y \|_F^2 + \frac{1-\alpha}{\alpha} \| \mathbf{\Gamma} \mathbf{W}_y \|_F^2. \end{aligned} \quad (9)$$



$$f(\mathbf{X}, \mathbf{Y}_{tr}; \mathcal{W}) = \mathbf{P} \mathbf{X} \mathbf{W}_x + (\mathbf{P} - \mathbf{C}) \mathbf{Y}_{tr} \mathbf{W}_y$$

exact predictor that minimally resolves label leakage

Cross-Entropy Loss

Theorem 2 *Under the same conditions as in Theorem 1 and Corollary 1, if we replace the MSE loss with categorical cross-entropy we obtain the bound*

$$\frac{1}{1 - \alpha_{splits}} \mathbb{E} \left[\text{CE}_{\mathcal{D}_{out}}(\mathbf{Y}_{out}, \mathbf{P}\mathbf{X}\mathbf{W}_x + \mathbf{P}\tilde{\mathbf{Y}}_{in}\mathbf{W}_y) \right] \geq \text{CE}_{\mathcal{D}_{tr}}(\mathbf{Y}_{tr}, \mathbf{P}\mathbf{X}\mathbf{W}_x + (\mathbf{P} - \mathbf{C})\mathbf{Y}_{tr}\mathbf{W}_y), \quad (11)$$

where $\text{CE}_S(\cdot, \cdot)$ denotes the sum of row-wise cross-entropy of S .

Nonlinear GNNs

Theorem 3 *Let $f_{GNN}(\mathbf{X}, \mathbf{Y}; \mathcal{W})$ denote an arbitrary GNN model with concatenated inputs $\mathbf{X}$ and $\mathbf{Y}$, and $\ell(\mathbf{y}, \hat{\mathbf{y}})$ a training loss such that $\sum_{i \in \mathcal{D}_{out}} \ell(\mathbf{y}_i, f_{GNN}[\mathbf{X}, \mathbf{Y}_{in}; \mathcal{W}]_i)$ is bounded for all $\mathcal{D}_{out}$. It then follows that*

$$\lim_{\alpha \rightarrow 1} \left\{ \frac{1}{1 - \alpha} \mathbb{E}_{splits} \left[\sum_{i \in \mathcal{D}_{out}} \ell(\mathbf{y}_i, f_{GNN}[\mathbf{X}, \mathbf{Y}_{in}; \mathcal{W}]_i) \right] \right\} = \sum_{i=1}^m \ell(\mathbf{y}_i, f_{GNN}[\mathbf{X}, \mathbf{Y}_{tr} - \mathbf{Y}_i; \mathcal{W}]_i).$$



Applications

Broader Applications

01

Trainable Label Propagation

02

Deterministic Label Trick Applied to GNNs

03

Trainable Correct & Smooth

Thank you for listening!